

AutoSchema: Self-Prompted Schema Induction and Evidence-Grounded Extraction for Materials Science Literature

Mingfang Zhu
Department of Computer Science &
Engineering
Washington University in St. Louis
St. Louis, MO, USA
mingfangz@wustl.edu

Yixin Chen
Department of Computer Science &
Engineering
Washington University in St. Louis
St. Louis, MO, USA
ychen25@wustl.edu

Zhiling Zheng
Department of Chemistry
Washington University in St. Louis
St. Louis, MO, USA
z.z@wustl.edu

Abstract

Scientific papers in materials science contain critical *experimental details* (e.g., reagents, synthesis conditions, and measured properties) for building structured databases and enabling downstream analysis, but large-scale structured extraction remains difficult. Classic *rule-based* systems rely on hand-written patterns and predefined targets, which can be brittle when high-value fields are domain-specific or shift across subdomains. Recent *LLM-based* extraction is more flexible, but it often still depends on manual *schema* and prompt design, where a schema specifies the target fields to extract into a table. We present **AutoSchema**, an iterative literature-mining pipeline that bootstraps both **retrieval and schema induction** from a small set of seed papers, then freezes the induced schema for scalable, evidence-grounded extraction. AutoSchema is not fixed-field extraction with a manually specified field list (e.g., ChatExtract-style [14] prompting), not entity/ontology-style recognition and normalization (e.g., MatScholar [21]), and not rule-based pattern engineering. Instead, it targets scalable literature mining under schema uncertainty, where the schema must be induced, stabilized, reused, and audited. Starting from 3–5 seed PDFs, AutoSchema expands a domain corpus via multi-source retrieval, induces and refines a compact schema set across rounds, and performs schema-conditioned, evidence-grounded extraction with JSONL audit logs and flattened tables. Across five independent seed sets, lightweight canonicalization and semantic normalization substantially improve schema agreement (mean pairwise Jaccard similarity 19.97%→43.41%; field-level overlap 32.50%→59.91%). In a *covalent organic frameworks (COF)* case study, Phase I expands the literature to a 1,656-paper corpus and Phase II applies the frozen schema to a 200-paper sample. Compared to ChemDataExtractor [11, 18], AutoSchema improves coverage and yields more verifiable extractions; compared to an evidence-grounded ChatExtract-style baseline [14] with a manually specified field list, AutoSchema provides stronger evidence support, while fixed-field prompting can reach higher fill rate and domain-specific coverage when the target schema is known. Additional small-scale pilot tests run the freeze-and-reuse pipeline on MOF/ZIF and Alzheimer’s disease amyloid/APP processing, and query-generation/retrieval tests on prompt-engineering

literature, illustrating AutoSchema’s transferability while keeping COF as the main benchmark.

CCS Concepts

• **Computing methodologies** → **Information extraction**; • **Information systems** → *Information retrieval*; • **Applied computing** → Chemistry.

Keywords

scientific information extraction, schema induction, information retrieval, literature mining, large language models, materials science, chemistry

ACM Reference Format:

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Resource Availability:

The source code of this paper has been made publicly available at <https://doi.org/10.5281/zenodo.20387054>. The corresponding GitHub repository is available at <https://github.com/lisazhuuu/AutoSchema>.

1 Introduction

Scientific papers in materials chemistry report key experimental details (e.g., reagents, synthesis conditions, and measured properties). Turning these details into structured tables is essential for building searchable databases and enabling downstream analysis. However, large-scale extraction remains difficult for two reasons. First, high-value targets are often *domain-specific* and vary across material families—for example, covalent organic frameworks (COFs), metal–organic frameworks (MOFs), and zeolitic imidazolate frameworks (ZIFs) emphasize different synthesis parameters and characterization conventions. Second, existing extraction paradigms do not scale well: *rule-based* pipelines require substantial manual schema mapping and pattern engineering and therefore have limited coverage when reporting styles change. On the other hand, *LLM-based* extraction is flexible but typically assumes a manually specified *schema* (a predefined list of target fields with definitions) and prompt, which is costly to design and hard to transfer across subdomains. This paper addresses a core bottleneck: *schema and prompt engineering do not scale*. In real life, people often cannot assume that target fields are known in advance, nor that a single



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prompt will transfer across venues and writing styles. Moreover, schemas induced from small seed sets can be unstable, which can lead to inconsistent downstream extraction. Our key question is: *can a system bootstrap both literature retrieval and schema construction from the literature itself, then reuse a frozen schema for scalable extraction?* We present AUTO SCHEMA, a domain-constrained, self-prompted iterative pipeline that learns what to extract before extracting it. Figure 1 gives a compact example of the two-phase output: raw paper text is converted into a frozen schema and then into evidence-grounded extraction outputs. Starting from a small set of seed PDFs, AUTO SCHEMA expands a domain corpus via multi-source retrieval, controls cross-round overlap via deduplication and top-off, induces and refines candidate fields across rounds, and freezes a reusable schema/prompt. Phase II then performs schema-conditioned passage retrieval and multi-value evidence-grounded extraction, producing JSONL audit logs and flattened tables for downstream analysis.

Why KDD should care. Real-world text-to-table mining faces schema shift and cold-start schema uncertainty—the “right” fields drift across corpora and over time, so pipelines cannot assume a fixed target schema. AUTO SCHEMA offers a reusable pattern: bootstrapped retrieval, schema induction/stabilization, freeze-and-reuse extraction, and evidence logging for auditability. We use COF literature as the main benchmark and include additional transfer tests to evaluate AUTO SCHEMA. Across five independent seed sets, lightweight canonicalization and semantic normalization substantially improve schema consistency (Jaccard similarity 19.97%→43.41%; field overlap 32.5%→59.91%). Phase I expands the COF literature to 1,656 papers; Phase II applies the frozen schema set to a 200-paper sample and compares against ChemDataExtractor and ChatExtract under a shared evaluation protocol, showing improved coverage and stronger evidence support. Additional small-scale pilot tests run the freeze-and-reuse pipeline on MOF/ZIF and Alzheimer’s disease amyloid/APP processing, and query-generation/retrieval tests on prompt-engineering literature.

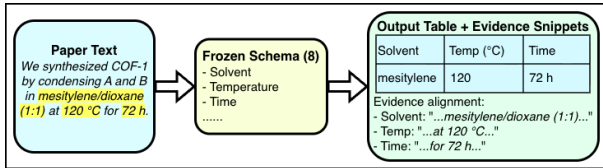


Figure 1: Illustrative example of AUTO SCHEMA: raw paper text → frozen schema → evidence-grounded extraction outputs.

Contributions. Our main contributions are:

- AUTO SCHEMA: an iterative pipeline that bootstraps retrieval and schema construction from seed papers and freezes a reusable schema/prompt for extraction.
- A schema stabilization procedure (canonicalization + semantic normalization) that significantly improves cross-seed consistency.
- Evidence-grounded extraction with audit logs and flattened tables, improving coverage and verifiability over rule-based baselines (e.g., ChemDataExtractor).

2 Related Work

Rule-based and NLP pipelines. Rule-based or classical NLP pipelines that target patterns or entities are common for mining materials papers [6, 7]. For example, ChemDataExtractor [11, 18] can extract structured information from chemistry text, but works best when target fields are known in advance and well covered by rules [4, 11, 18]. When fields become more domain-specific or shift across subdomains (e.g., COF vs. MOF vs. ZIF), reusable schemas and downstream database mapping become harder [17]. ChemDataExtractor 2.0 [11] moves toward more structured representations (e.g., ontology-style organization), but still largely assumes the schema is designed before extraction rather than induced from the literature. MatScholar [21] builds cleaner and more searchable corpora through materials entity recognition and normalization, but targets entity-level extraction rather than task-driven field tables with adaptable schemas. Closer to synthesis-focused mining, the synthesis recipes pipeline by Kononova et al. [5] demonstrates large-scale synthesis extraction, while the extraction target is still pre-specified and it does not directly address stabilizing schemas across small-seed runs.

LLM-based extraction and evidence support. Recent systems use LLMs to improve extraction robustness across diverse scientific writing styles [1, 13]. ChatExtract [14] is a materials-focused, evidence-grounded extraction method. In our comparison, it is best viewed as a strong fixed-schema reference rather than a fully symmetric schema-induction baseline, because it assumes manually specified target fields and focuses on extracting values for predefined fields. Relatedly, domain-specific LLM pipelines extract synthesis details in reticular materials [19]; they can be effective in focused settings but still rely on human-designed prompts and fixed structures. Together, these works highlight the same bottleneck we noted in the introduction: *prompt and schema engineering do not scale* [10].

3 Task & Problem Formulation

We formulate the main benchmark as large-scale information extraction from materials science literature, focusing on experimental and synthesis details. Additional artifact tests later examine transfer beyond this setting. The key challenge is that there is no single “gold” schema fixed beforehand: schemas shift across subdomains (e.g., COF/MOF/ZIF), venues, and writing styles, and users often do not know the full target schema in advance. Thus, the bottleneck is not only extracting values, but also deciding *what* to extract and making schemas reusable.

Input. AUTO SCHEMA takes a per-run seed set of 3–5 PDF papers. It converts PDFs to text, generates structured search queries, expands the corpus with source-specific query adaptation, induces a schema set, and later extracts values.

Output. AUTO SCHEMA outputs: (1) a retrieved corpus; (2) a schema set $\mathcal{S}$ in human-friendly and machine-readable forms; (3) schema-conditioned extraction outputs, where each field may contain one or more values with supporting evidence; and (4) JSONL audit logs plus flattened CSV tables, with missing values marked as N/A and missing evidence marked by a placeholder (e.g., NO EVIDENCE FOUND).

Problem setting. Given a small seed set, the system should (i) bootstrap retrieval to build a larger, diverse corpus, (ii) induce and update a schema set from retrieved papers, and (iii) finalize a schema set for scalable reuse. Ideally, the final schema should be stable across independent runs from different seed sets; otherwise, downstream extractions become inconsistent.

Evaluation metrics. We evaluate AUTOSchema from five angles: (i) **Schema stability**, similarity across schema sets induced from independent seed sets; (ii) **Overall fill rate**, the average per-field fill rate on a paper sample; (iii) **Evidence match**, evaluated on a fixed audited subset using an LLM-based judge; (iv) **Domain-specific coverage**, field-level coverage of a predefined checklist via canonical alias matching; and (v) **Manual spot-check**, a small human audit of extracted values, evidence, and source text.

4 Method

4.1 System Overview

AUTOSchema is a two-phase pipeline for literature mining under schema uncertainty: it learns what to extract before extracting it. Each run starts from a small seed set of 3–5 PDFs and produces: (i) a frozen schema set with field names, definitions, aliases/examples when available, (ii) schema-conditioned extraction outputs, and (iii) audit logs that connect extracted values to supporting evidence. The pipeline follows a freeze-and-reuse workflow. **Phase I** bootstraps a domain-focused corpus from seed papers and iteratively induces a compact schema from retrieved papers. **Phase II** reuses the frozen schema without adding new fields, retrieves schema-relevant passages from each paper, and performs multi-value evidence-grounded extraction. The final outputs are JSONL audit logs and flattened CSV tables for downstream analysis. Compact prompt constraints, domain/source configurations, and Phase II output examples are provided in Appendix A.

4.2 Phase I: Retrieval Bootstrapping and Schema Induction

Phase I is organized around three higher-level stages: seed-conditioned retrieval, local candidate generation, and schema refinement/freezing. This presentation abstracts away low-level implementation mechanics while keeping the core iterative schema-induction loop clear.

4.2.1 Seed-conditioned query generation and retrieval. Given 3–5 seed papers, the system first converts each PDF into text and applies a domain-specific extraction prompt to produce a compact structured snapshot. This snapshot summarizes metadata and domain signals useful for search, and is used to condition query generation rather than relying on a manually written query list. Query generation is domain-constrained through DomainSpec. The specification defines the domain boundary, last-tier focus, aliases, prompts, and source-specific query preferences. From the seed snapshots, the system generates structured queries containing Boolean and natural-language forms, short synonyms, and a brief rationale. Queries are validated against the domain/last-tier focus, normalized, and deduplicated before retrieval. Most domain customization is isolated in DomainSpec: retargeting from COF to MOF/ZIF or to another supported domain mainly changes the domain specification

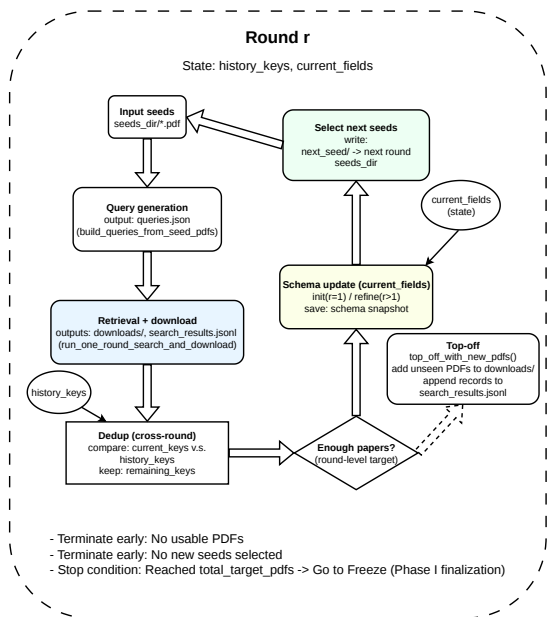
and source configuration, while the retrieval and schema-induction logic remains shared. Table 1 summarizes the tested domain specifications and retrieval sources used by the current implementation, and Appendix A.1 gives a compact prompt/query example. For each round, the retrieval driver adapts queries to each source. The artifact uses repo-friendly sources: chemistry runs use arXiv, OpenAlex, Nature, and CORE; Alzheimer’s disease runs use PubMed, bioRxiv, OpenAlex, and CORE; and CS prompt-engineering runs use arXiv and OpenAlex. ACS/browser-based fallback from the submitted version is removed to avoid institution-dependent behavior in the public artifact. arXiv retrieval uses polite delay, small page size, retries, and a circuit breaker: if repeated 429/503 failures occur, arXiv is disabled for the rest of the run and the pipeline continues with the remaining sources. OpenAlex/Crossref-style requests may include contact information when available [2, 15]. Retrieved candidates from different sources and queries are merged using a record key with priority DOI → source ID → normalized title. This prevents the same paper from being counted multiple times across sources or rounds. Candidate papers are then ranked toward domain relevance and downloadability, downloaded when possible, and cleaned by removing invalid PDFs or empty directories.

4.2.2 Local candidate generation and schema refinement. After retrieval, AUTOSchema uses the retained PDFs to update the schema. Before schema design, the implementation applies relevance filtering so that off-topic or low-signal papers are less likely to dominate field induction. The remaining papers are sampled for local candidate generation: each paper proposes reusable extraction fields with short descriptions, examples, and evidence snippets when available. Candidate fields are then merged into canonical field candidates and selected according to support, evidence quality, and domain relevance. In the first round, the system initializes a schema from the sampled papers. In later rounds, it refines the previous schema using newly retrieved papers: supported fields can be strengthened, weak or overly generic fields can be dropped, and new supported fields can be added. This iterative update lets the schema reflect the expanding corpus while still keeping the schema compact. The current implementation also applies a final semantic-duplicate merge before freezing the schema. This pass merges fields that capture the same underlying concept, keeps one representative field per concept, and removes overly broad fields when more specific fields already cover the information. This is separate from the stability analysis in §5: canonicalization and semantic normalization are used to compare schema sets across runs, while Phase II extraction uses the frozen schema produced for that run.

4.2.3 Round orchestration and schema freezing. AUTOSchema runs Phase I as a round-based loop controlled by an orchestrator. Figure 2 summarizes this round-level control flow before final schema freezing. Each run maintains two persistent states: `history_keys`, the set of unique records accepted so far, and `current_fields`, the current schema state. In each round, the orchestrator builds queries from the current seeds, retrieves and downloads papers, deduplicates against `history_keys`, optionally tops off the round with unseen papers, updates `current_fields`, and selects new seeds for the next round. Seed refresh helps the system explore nearby literature while staying on topic. The orchestrator scores candidate

Table 1: Compact domain specifications and retrieval sources used by the artifact.

Domain	Tested focus	Domain boundary / last-tier role	Retrieval sources
chem	COF, MOF, ZIF	Materials chemistry / porous-material subdomains; the last tier controls the target material family and query focus.	arXiv, OpenAlex, Nature, CORE
ad	Amyloid / APP processing	Alzheimer’s disease biomedical literature; the last tier keeps queries focused on amyloid production, APP processing, and related mechanisms.	PubMed, bioRxiv, OpenAlex, CORE
cs	Prompt engineering	Computer-science literature; tested for query generation and retrieval rather than full schema induction or Phase II extraction.	arXiv, OpenAlex

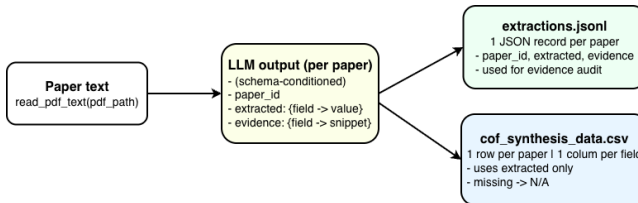
**Figure 2: Round-based orchestration in Phase I, organized as seed-conditioned retrieval, schema refinement, and seed refresh before final schema freezing.**

papers using similarity to the current seeds and tier-keyword signals, filters out low-signal candidates, and writes selected PDFs as the next round’s seeds. The run stops if a round yields no usable PDFs, if no new seeds can be selected, or if the target number of accepted PDFs is reached. After the loop finishes, the final schema state is frozen into a machine-readable field set. This frozen schema is the only schema used for Phase II within the run; Phase II does not unfreeze the schema or add new fields midstream. The system compiles the frozen schema into the extraction prompt and saves both the schema and prompt for reuse.

4.3 Phase II: Multi-value evidence-grounded extraction

Phase II performs schema-conditioned extraction over a PDF corpus using the frozen schema from Phase I. For each paper, the extractor outputs a JSON object containing paper_id and extractions, where each field in extractions follows {values:

[...], evidence: [...]}. Evidence entries include supporting quotes and passage references. Appendix A.2 shows the compact output format and a representative COF schema excerpt. The extractor first loads the frozen schema and builds a system prompt from the schema fields, descriptions, aliases, and example evidence. Each PDF is converted to text using pypdf [16]. Instead of sending the full paper directly, the system performs schema-conditioned passage retrieval: it scores paragraphs using field names, aliases, member field names, and evidence-snippet overlap, then sends only the selected passages to the LLM. This reduces irrelevant context and makes evidence grounding more explicit. The LLM is instructed to extract only values supported by the provided passages, return empty lists when no value is stated, and keep value strings separate from evidence quotes. Fields with array/enum types may contain multiple distinct values; single-valued fields are normalized to one value during post-processing. Robust JSON parsing and normalization handle common formatting issues, legacy output shapes, duplicate values, and overly long evidence strings. The full normalized record is appended to extractions. jsonl for auditing. In parallel, the extracted values are flattened into schema_values.csv, with one row per paper and one column per field. Empty value lists are written as N/A, and multi-values are joined with a delimiter. Figure 3 summarizes the JSONL audit log and flattened CSV outputs.

**Figure 3: Phase II outputs: JSONL audit records with multi-value extractions and evidence, plus a flattened CSV table (N/A for missing values).**

5 Experiments

5.1 Data & Setup

Main COF benchmark. Our main benchmark is covalent organic framework (COF) literature. We first prepare a 300-paper seed pool by running Phase I in retrieval-only mode from 3 seed PDFs, then construct 5 independent seed sets with 5 randomly selected papers. We verify that no paper appears in more than one seed set. For each

seed set, we run the round-based Phase I orchestration pipeline to collect papers for schema induction. In the main setup, we use up to 3 rounds, 10 queries per round, 40 results per source, 8 seeds for the next round, an overlap threshold of 0.15, a target of 150 PDFs, and a minimum continuation threshold of 5 retained PDFs. Each run outputs one frozen schema set. We then compare the five frozen schema sets using Jaccard similarity and field-level overlap. Phase I expansion yields a COF corpus pool of 1,656 papers. For controlled Phase II evaluation, we sample a fixed 200-paper subset from this pool and run extraction on the same subset for all five frozen schema sets and both baselines.

Seed-refresh ablation. We ablate the Phase I *seed refresh* mechanism by rerunning Phase I with identical hyperparameters but keeping the seed set fixed across rounds (no-refresh). As shown in Appendix C.1, seed refresh increases late-round expansion: the number of *new unique* papers rises from 44/29 in Rounds 2/3 under no-refresh to 52/48 with refresh for Set01, yielding a larger expanded corpus after Round 3 (233 $\rightarrow$ 260 unique papers). We further compare the induced no-refresh schema to refreshed runs using exact column-name overlap. The overlap is low (Jaccard 0.067–0.143, with only 1–2 shared columns out of 8; Table 2), indicating that even when corpus growth differences are modest, schema induction can drift under different exploration trajectories and naming choices.

Table 2: Schema similarity between refreshed runs and the no-refresh run.

Run	F	Overlap	Jaccard	Overlap/No-refresh
Set_01	8	1	0.067	0.125
Set_02	8	2	0.143	0.250
Set_03	8	2	0.143	0.250
Set_04	8	1	0.067	0.125
Set_05	8	2	0.143	0.250

Additional transfer checks. Beyond the main COF benchmark, we conduct smaller transfer checks on MOF, ZIF, Alzheimer’s disease amyloid/APP processing, and CS prompt-engineering literature. The MOF/ZIF and AD checks run the freeze-and-reuse pipeline at small scale, including Phase II pilot extraction, while the CS check covers query generation and retrieval only. These checks are smaller than the COF benchmark and are used to assess portability rather than replace the main evaluation.

5.2 Baselines

We compare AUTOSchema against two representative baselines: ChemDataExtractor, a rule-based/NLP extraction toolkit, and a ChatExtract-style LLM extraction workflow. These baselines cover traditional chemistry text mining and evidence-grounded LLM extraction, but they do not make the same assumptions as AUTOSchema: CDE does not induce our task schema, and ChatExtract-style extraction is best viewed as a strong fixed-schema reference rather than a fully symmetric schema-induction baseline.

ChemDataExtractor (CDE). Our first baseline is ChemDataExtractor [11, 18]. We apply it directly to the same 200 PDFs and extract several synthesis-related signal types with sentence-level evidence: (i) general chemical entities (CEMs), (ii) temperatures, (iii) reaction times/durations, (iv) solvent-related entities detected from synthesis-context sentences, and (v) BET surface-area and pore-size mentions. The appendix per-field table reports the 196 papers with valid CDE-parsed outputs. To improve precision, we apply lightweight heuristics implemented in our baseline script, including synthesis-context gating for temperature/time, range-based plausibility filters, and keyword cues to suppress non-synthesis false positives such as figure-panel references or microscopy scale-bar sentences. Each extracted mention is stored with a short evidence sentence for downstream auditing.

ChatExtract-style LLM extraction. The other baseline is ChatExtract, an LLM-based extraction workflow that focuses on strict evidence grounding [14]. We implement a ChatExtract-style script with a three-step pipeline: (i) *identification* retrieves short verbatim evidence spans from PDF text chunks, using a keyword-based chunk pre-filter to reduce cost; (ii) *extraction* predicts the target value using only the retrieved spans and selects evidence from the span set; and (iii) a *follow-up verifier* confirms or revises the value while still being restricted to the same evidence spans. We also apply an automatic traceability check requiring the reported evidence to match a normalized substring of an aligned chunk; otherwise, the output is labeled as evidence-not-in-text. Unlike AUTOSchema, this baseline does not induce a schema and is directly prompted with predefined COF-critical target fields.

Common evaluation protocol. All methods are evaluated on the same 200-paper subset under the same missing-value convention (N/A). For coverage metrics, we use each method’s extracted table or mapped mention outputs. For evidence-based evaluation, we use each method’s stored evidence snippet or sentence-level evidence and apply the evidence-supported extraction metric described in Section 5.3.

5.3 Metrics

Phase I: schema stability. We evaluate schema stability across the $K=5$ frozen schema sets induced from independent seed sets. Before comparison, we lightly normalize schema fields to reduce superficial naming variation, including canonicalization (lowercasing, removing unit suffixes such as `_c/_m2g`, synonym replacement, and unifying underscores/spaces). This lets us compare field semantics rather than relying only on surface names.

Jaccard similarity. Let S_i denote the normalized schema set from seed set i . We compute pairwise Jaccard similarity

$$J(S_i, S_j) = \frac{|S_i \cap S_j|}{|S_i \cup S_j|},$$

and report the mean Jaccard score over all $\binom{K}{2}$ pairs. Higher Jaccard similarity indicates stronger agreement across independent seed sets.

Field-level overlap. We also evaluate a field-level overlap rate, which measures how many fields are consistently present across runs. For each normalized field in the union $\bigcup_{i=1}^K S_i$, we count how

many schema sets contain that field. Fields appearing in at least 80% of runs (i.e., ≥ 4 out of 5) are marked as shared. The field-level overlap rate is the proportion of fields in the union that are shared. This complements Jaccard by emphasizing consensus fields that remain stable across different runs.

Phase II: extraction quality at scale. On the 200-paper subset used for all Phase II evaluations, we use metrics that reflect coverage, evidence grounding, and domain-relevant coverage. Missing values are represented as N/A; for multi-value outputs, a field is considered filled if at least one non-N/A value is produced.

Field fill rate. We measure how completely a method fills the extraction table. Over N papers and $|F|$ target fields, the overall fill rate is

$$\text{FillRate} = \frac{\#\{\text{non-N/A cells}\}}{N \cdot |F|}.$$

We report both per-field fill rates and the overall average across all fields.

Evidence-supported extraction. This metric evaluates whether an extracted value is supported by the quoted evidence from the paper text. We randomly sample 20 papers, evaluate all non-N/A fields with evidence for these papers, and use GPT-4o [12] to judge whether each extracted value matches its provided evidence snippet (match=1, otherwise 0). We reuse the same 20-paper subset across all schema sets and baselines for fairness, and report the mean match rate over the resulting (paper, field) records. Missing values or NO EVIDENCE FOUND are counted as mismatches for this metric.

Domain-specific field coverage. Some fields are especially important for COF-focused synthesis and characterization. We define a COF-critical checklist F_{domain} containing *Building blocks*, *Linkage*, *PXRD pattern*, *FT-IR*, *Crystallinity*, *Solvent stability*. Because different methods produce outputs with different structures, we compute domain-specific coverage in a method-appropriate way. **(i) Fixed-field prompting (ChatExtract-style).** Since the domain fields are explicitly enforced as extraction targets, we report the restricted fill rate over F_{domain} :

$$\text{DomainCoverage}_{\text{fill}} = \frac{\#\{\text{non-N/A cells in } F_{\text{domain}}\}}{N \cdot |F_{\text{domain}}|}.$$

(ii) Schema-based / mention-based outputs that require mapping. For AUTOSchema and CDE, the outputs do not necessarily use the checklist field names directly. We therefore report mapped domain coverage based on canonical alias matching to the checklist categories. A domain category is counted as covered for a paper if at least one extracted field/value or mention can be mapped to that category (details in Appendix B.2). This distinction matters: restricted fill rate measures how often explicitly requested COF-critical fields are filled, while mapped coverage measures whether induced or mention-level outputs can be aligned to the checklist through post-hoc normalization.

Manual spot-check. To complement the LLM judge, we manually audit sampled extractions from four representative fields. Each instance is labeled Correct, Partial, or Unsupported by comparing the extracted value, logged evidence, and source text. *Correct* means the value is supported by the source; *Partial* means the value is grounded but incomplete, underspecified, or not fully normalized;

and *Unsupported* means the value is not backed by the cited text. If the paper does not contain the information and the system outputs N/A, we count it as Correct.

5.4 Main Results

Phase I: schema stability. Across the $K=5$ independent COF runs, schema induction shows substantial variance when evaluated using raw field names. After canonicalization and semantic normalization (e.g., lowercasing, removing unit suffixes, synonym replacement, and unifying underscores/spaces), agreement between frozen schema sets improves substantially: mean pairwise Jaccard similarity increases from 19.97% to 43.41%, and field-level overlap increases from 32.5% to 59.91%. This suggests that a large portion of cross-run “instability” comes from superficial naming differences rather than semantic disagreement. Tables 3 and 4 summarize the pairwise Jaccard and field-overlap results.

Table 3: Pairwise Jaccard before $\rightarrow$ after normalization.

Pair	Set_02	Set_03	Set_04	Set_05
Set_01	23.08 $\rightarrow$ 75.00	23.08 $\rightarrow$ 36.36	14.29 $\rightarrow$ 40.00	33.33 $\rightarrow$ 40.00
Set_02	–	14.29 $\rightarrow$ 50.00	33.33 $\rightarrow$ 40.00	6.67 $\rightarrow$ 40.40
Set_03	–	–	14.29 $\rightarrow$ 36.36	23.08 $\rightarrow$ 36.36
Set_04	–	–	–	14.29 $\rightarrow$ 40.00

Table 4: Per-run field overlap before vs. after normalization.

Run	Before	After	Δ	Fields
Set_01	37.50	64.29	+26.79	8
Set_02	31.25	67.86	+36.61	8
Set_03	31.25	53.12	+21.87	8
Set_04	31.25	57.14	+25.89	8
Set_05	31.25	57.14	+25.89	8

We also re-test three schema sets with a fixed decoding seed (seed= 42) to reduce generation stochasticity. This test still shows lower stability (mean pairwise Jaccard 23.57%, overlap 37.50%), indicating that schema induction remains sensitive to retrieval and corpus composition even when generation randomness is reduced.

Phase II: extraction at scale under frozen schemas. We evaluate extraction quality on the same 200-paper subset across all five frozen schema sets. The overall field fill rate ranges from 56.73% to 62.11%, with an average fill rate of 59.81%. Using the shared 20-paper evidence subset and the evidence-supported metric described in Section 5.3, the evidence-supported rate is consistently high across schema sets, ranging from 78.79% to 84.69%. Table 5 reports the per-schema summary. For baseline comparison, we use Set01, whose fill rate is closest to the five-schema mean.

Table 5: AUTOsCHEMA Phase II summary across five frozen schema sets.

Schema Set	Fill Rate	Evidence Acc.
Set_01	59.81%	83.52%
Set_02	62.11%	81.33%
Set_03	58.79%	78.79%
Set_04	61.60%	84.69%
Set_05	56.73%	81.40%

Table 6 reports the per-field fill rates for Set01. The field-level rates range from 49.0% to 69.0%, with higher fill rates for building-block and solvent-related fields than for BET-specific surface-area fields.

Table 6: Per-field fill rate for the induced schema Set_01 ($N = 200$ papers).

Field Name	Filled	Missing	Fill %
monomer_building_blocks	138	62	69.0
building_block_identities	136	64	68.0
solvent_system	134	66	67.0
reaction_temperature_c	121	79	60.5
linkage_chemistry	118	82	59.0
reaction_time_h	112	88	56.0
specific_surface_area_bet	100	100	50.0
bet_surface_area_m2_g	98	102	49.0
Global Avg.	–		59.81

Manual spot-check results. To complement the LLM-based evidence judge, we manually audit 32 sampled instances: 8 samples each for solvent_system, bet_surface_area_m2_g, building_block_identities, and monomer_building_blocks. As shown in Table 7, 26/32 instances are Correct, 6/32 are Partial, and no Unsupported cases are observed. Partial cases mostly arise from ambiguous or incomplete reporting, such as solvents appearing in preparation or characterization contexts rather than clearly as synthesis solvents, instead of unsupported hallucination.

Table 7: Manual spot-check of extracted values, evidence, and source text.

Field	Correct	Partial	Unsupported
solvent_system	5	3	0
bet_surface_area_m2_g	8	0	0
building_block_identities	6	2	0
monomer_building_blocks	7	1	0
Total	26	6	0

Transfer checks beyond COF. The smaller transfer checks support portability but also show that performance depends on retrieval precision, schema granularity, and domain specificity. MOF is the strongest chemistry transfer case: the pipeline runs through schema induction and Phase II pilot extraction, with useful application, linker, metal-node, and surface-area fields. ZIF also runs through the same freeze-and-reuse workflow, but retrieval drifts more often

toward broader MOF/porous-material papers and some induced fields are narrower, so its extraction completeness is lower than the main COF benchmark. AD amyloid/APP processing demonstrates preliminary cross-domain transfer: query generation and retrieval are meaningful, the induced schema includes amyloid/APP-relevant fields, and a small Phase II smoke test produces reasonable biomedical terms, although its fill rate is lower than COF because the schema is narrower and some retrieved papers are broader AD papers. CS prompt engineering is only a query-generation/retrieval check and should not be interpreted as a full benchmark. Table 8 summarizes representative fields or outputs from these smaller transfer checks.

Table 8: Representative transfer checks beyond the main COF benchmark.

Setting	Tested scope	Representative fields / outputs
chem/MOF	Small pipeline pilot	full- application_area, metal_node_composition, organic_linker, surface_area
chem/ZIF	Small pipeline pilot	full- application_domain, characterization_methods, synthesis_method, crystallinity
ad/amyloid	Small pipeline pilot	full- app_processing_pathway, amyloidogenic_secretases, amyloid_species, cell_model
cs/prompt	Query/retrieval only	Meaningful prompt-engineering queries and retrieved CS papers; schema induction and Phase II were not tested.

Domain-specific coverage. Overall fill rate is influenced by broadly applicable fields (e.g., general synthesis conditions), so we additionally evaluate coverage on COF-critical categories. The COF-critical checklist was derived from a small set of 8 closely read COF papers and consolidated through synonym grouping [3, 8, 9, 20, 22–25]. In our experiments, the domain-specific categories are *Building blocks*, *Linkage*, *PXRD pattern*, *FT-IR*, *Crystallinity*, *Solvent stability*. We treat them as domain-specific because they are closely tied to COF identity and validation and are often expressed in diverse narrative forms rather than a consistent value-plus-unit pattern. The remaining four categories, *Synthesis method*, *SBET*, *Pore size*, *Pore volume*, are treated as more general materials-property signals that commonly appear outside COF reporting. For Set01, AUTOsCHEMA maps to 2/6 COF-critical categories (33.33%), mainly *Building blocks* and *Linkage*. Over all ten categories, it maps to 3/10 categories (30.0%), adding the general surface-area category *SBET*. Appendix B.1 reports the per-category mapping results.

Baseline comparison. To contextualize AUTOsCHEMA, we compare it against CDE and ChatExtract-style extraction on the same 200-paper subset using the same three metrics: overall fill rate, evidence-supported accuracy, and domain-specific field coverage. Table 9 summarizes the compact comparison.

Table 9: Performance comparison of three methods on the 200-paper subset.

Method	Fill Rate	Evidence Acc.	Domain-Specific Cov.
Set01 (AuTo)	59.81%	83.52%	33.33%
CDE (Base. A)	42.16%	54.24%	16.67%
ChatExt. (Base. B)	63.35%	35.00%	71.58%

Baseline A: ChemDataExtractor (CDE). CDE is a rule-based/statistical chemistry text-mining toolkit that extracts broadly available signals such as temperature, duration, solvents, quantities with units, and general chemical entities. On the 200-paper subset, CDE achieves an overall fill rate of **42.16%**. For evidence grounding, we evaluate CDE on the **shared 20-paper evidence set**, where it attains an evidence-supported accuracy of **54.24%**. Its domain-specific coverage is **16.67%**. Since CDE does not induce or output our task-specific COF schema, we apply lightweight post-hoc mapping from CDE outputs to our target field categories, selecting the first reasonable non-obviously-wrong candidate when multiple values are available.

Baseline B: ChatExtract-style LLM extraction. The ChatExtract-style baseline uses LLM extraction with strict evidence grounding, but it does not learn or construct a schema. Instead, we directly prompt it to return values for predefined target field categories. On the same 200-paper subset, ChatExtract-style extraction achieves an overall fill rate of **63.35%**. On the **shared 20-paper evidence set**, it attains an evidence-supported accuracy of **35.00%**. Its domain-specific coverage is **71.58%**, computed as the restricted fill rate over the six COF-critical fields because these fields are directly enforced at inference time. Thus, the ChatExtract-style domain-specific score should be read as a strong fixed-schema reference rather than a fully symmetric schema-induction comparison: it measures how often explicitly requested COF-critical fields are filled, not whether an induced schema can be mapped to those fields.

Summary. Compared with CDE, AuToSCHEMA improves coverage and evidence-supported accuracy while inducing a task schema from the corpus. Compared with ChatExtract-style extraction, AuToSCHEMA provides stronger evidence support, while fixed-field prompting reaches higher fill rate and COF-critical coverage when the target schema is given. These results support the intended use case of AuToSCHEMA: literature mining under schema uncertainty, where the schema must be induced, stabilized, reused, and audited.

6 Discussion and Limitations

Discussion. Our results reveal a practical “freeze-and-reuse” pattern for literature mining under schema uncertainty. When a schema set is induced from a corpus expanded from small seed papers, differences in surface forms can drive apparent discrepancies across runs, while canonicalization and semantic normalization recover a much higher level of agreement. At the same time, the seed=42 re-test indicates that schema induction remains sensitive to upstream retrieval and sampling, suggesting that stability is not only constrained by decoding randomness, but also by corpus composition and the extent of exploration in each round. In Phase II, evidence logging materially improves extraction verifiability by pairing extracted

values with supporting snippets or standardized missing-evidence markers, making the outputs easier to audit for downstream analysis. The transfer checks further suggest that the same overall design can be reused beyond the main COF benchmark, but with different levels of maturity. Chemistry transfer to MOF and ZIF is the closest setting because the corpus, terminology, and schema structure remain near the original materials domain; small Phase II pilots run through the freeze-and-reuse workflow, but extraction completeness remains below the main COF benchmark. AD amyloid/APP processing is a more distant biomedical test: the pipeline can run end-to-end and produce meaningful amyloid-related fields, but the Phase II fill rate is lower because the induced schema is narrower and some retrieved papers are broader AD papers. CS prompt-engineering literature is currently tested only at the query-generation and retrieval level, so it should be interpreted as an early retrieval demonstration rather than a full schema-induction benchmark.

Evaluation limitations. First, COF remains the only full formal benchmark in this paper. MOF/ZIF and AD amyloid/APP processing are smaller full-pipeline pilot checks, while CS prompt-engineering is limited to query generation and retrieval; none is an equally comprehensive benchmark to COF. Second, evidence-supported accuracy is measured on a fixed 20-paper evidence set and relies on an LLM-based judge. Although fairness is improved by reusing the same 20 papers across methods, the estimate can still be noisy, and the judge may have its own bias or failure modes, such as being overly permissive or overly strict on partial matches. The manual spot-check helps address extraction correctness more directly, but it covers only 32 sampled instances and is not a substitute for large-scale expert annotation. Third, domain-specific coverage is inherently definition-dependent. Our COF-critical list is based on a small set of 8 carefully read papers, and the resulting terms are grouped into categories through manual consolidation of synonyms. The target is therefore transparent and task-motivated, but it may not capture all COF-relevant attributes or reflect other subdomains. Moreover, some baseline measurements are not strictly comparable. For example, domain coverage can be framed as schema-field hits in schema-based or mention-based methods, as in AuToSCHEMA/CDE mapping; but in the ChatExtract-style baseline, the task is more naturally expressed as a restricted fill rate over a predefined and strictly enforced field list. This can lead to higher domain-specific scores because the COF-critical fields are explicitly provided at inference time.

Scope and future directions. AuToSCHEMA currently provides its strongest evidence on COF literature, with smaller transfer checks across additional chemistry, biomedical, and CS retrieval settings. Its behavior may change under different sources, seed characteristics, rate limits, or subdomain shifts. In particular, artifact-scale runs can differ from the reported formal experiments because source availability, arXiv rate limits, seed selection, and retrieval noise affect corpus composition. Accordingly, a natural next step is to identify which design decisions account for the largest share of variance and improvement: (i) Phase I expansion knobs such as query budget, per-source depth, seed refresh, and dedup/top-off thresholds; (ii) schema normalization and semantic-merge rules;

and (iii) Phase II controls such as passage retrieval, chunking, evidence selection, and verification. Expanding to more subdomains and adding larger expert-labeled audits would further strengthen conclusions about generality, faithfulness, and downstream usability.

7 Conclusion

We present AUTOSchema, an end-to-end pipeline for scalable literature mining under schema uncertainty. Starting from 3–5 seed PDFs, it (i) bootstraps retrieval to expand a domain corpus, (ii) induces and stabilizes a task schema from the expanded corpus, and (iii) performs freeze-and-reuse, evidence-grounded extraction with auditable outputs. Across independent COF runs, raw field names suggest substantial schema drift, but lightweight canonicalization and semantic normalization substantially improve agreement (mean pairwise Jaccard: 19.97%→43.41%; overlap: 32.5%→59.91%; Tables 3–4), indicating that much apparent instability is driven by surface-form variation rather than semantic mismatch. At the same time, a controlled re-test (seed=42) shows that schema induction remains sensitive to upstream retrieval and corpus composition beyond decoding randomness, highlighting a practical trade-off between exploration and schema reuse. In Phase II, schema-conditioned passage retrieval and evidence-grounded extraction improve verifiability by pairing extracted values with supporting evidence or standardized missing-evidence markers, enabling downstream users to inspect and filter extracted tables. The manual spot-check further supports this auditability result at small scale, with no unsupported cases observed among the sampled instances. Beyond the main COF benchmark, transfer checks on MOF/ZIF, AD amyloid/APP processing, and CS prompt-engineering retrieval suggest that the same design can be adapted across settings, while also showing that retrieval precision and domain specificity affect performance. Overall, our results clarify when schema-induced extraction is preferable to fixed-field prompting: when target fields are unknown, shifting across subdomains, or need to be reused over a growing corpus, AUTOSchema provides a reusable methodological pattern; when the schema is stable and narrowly defined, fixed-field prompting remains a strong reference.

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A Method Details and Implementation Examples

A.1 Prompt and Query Examples

Table 10: Compact prompt constraints and query example used in Phase I.

Component	Paper-facing summary
Seed-extraction prompt	Given seed-paper text, return JSON only. The output records bibliographic metadata and compact domain signals used for later query construction, such as material identity, synthesis conditions, measured properties, and short notes. Unknown fields are returned as null or empty lists; the model is instructed not to invent missing information.
Query-generation prompt	From seed snapshots and the active DomainSpec, generate a small set of structured search queries. Each query contains a Boolean form, a natural-language description, short synonyms, and a brief rationale. Queries must stay within the domain boundary and target the last-tier focus defined by the domain specification.
Source adaptation	The Boolean query is used when supported, while source-specific simplified forms are used for APIs that do not support full Boolean/arXiv-style syntax. The retrieval logic then merges candidates across sources and rounds using record-level deduplication.
Example query	<i>COF membrane fabrication:</i> Search for covalent organic framework membrane fabrication through interfacial, vapor-liquid, steam-assisted, layer-by-layer, or electrochemical routes, emphasizing scalable synthesis of thin crystalline COF films.

A.2 Phase II Output Format and Schema Excerpt

Phase II loads the frozen schema, retrieves schema-relevant passages from each paper, and returns JSON-only multi-value outputs. For each paper, the normalized record follows {"paper_id": "...", "extractions": {field: {values: [...], evidence: [...]}}}, where each evidence entry contains a short quote and passage reference. Full records are written to extractions.jsonl; values are flattened into schema_values.csv, with N/A for empty fields.

Representative COF schema fields include solvent system, reaction temperature c, reaction time h, bet surface area, building block identities, and linkage chemistry.

B Experiment Details

B.1 AutoSchema: Set 01 Per-Field Fill Rate & Domain-Specific Coverage

Table 11: Set01 domain-specific coverage summary. “Hit” indicates whether the domain field category can be mapped to at least one extracted schema field.

Field Category	Hit	Matched Fields
Building blocks	✓	building_block_identities, monomer_building_blocks
Linkage	✓	linkage_chemistry
Synthesis method	×	–
PXRD pattern	×	–
FT-IR	×	–
SBET	✓	bet_surface_area_m2_g, specific_surface_area_bet
Pore size	×	–
Pore volume	×	–
Crystallinity	×	–
Solvent stability	×	–
Coverage (all fields): $3/10 = 0.30$		
Coverage (domain-specific): $2/6 = 0.33$		

B.2 Baselines: Fill Rate and Domain Coverage

ChemDataExtractor (CDE).

Table 12: Field extraction statistics for CDE ($N = 196$ papers). Global average fill rate: 42.16%.

Field Name	Filled	Missing	Fill Rate (%)
building_block_identities	195	1	99.49
solvent_system	157	39	80.10
reaction_time_h	115	81	58.67
reaction_temperature_c	104	92	53.06
bet_surface_area_m2_g	90	106	45.92
specific_surface_area_bet	0	196	0.00
linkage_chemistry	0	196	0.00
monomer_building_blocks	0	196	0.00
Global Average	—		42.16%

Table 13: CDE domain-specific coverage summary for the induced schema (“Hit” indicates mapped categories).

Field Category	Hit	Matched Fields
Building blocks	✓	general_entities, synthesis_sentence_entities
Linkage	×	–
Synthesis method	×	–
PXRD pattern	×	–
FT-IR	×	–
SBET	✓	bet_surface_area
Pore size	✓	pore_size
Pore volume	×	–
Crystallinity	×	–
Solvent stability	×	–
Coverage (all fields): $3/10 = 0.30$		
Coverage (domain-specific): $1/6 = 0.17$		

ChatExtract (Baseline 2).

Table 14: ChatExtract fill rate summary across ten field categories ($N = 200$ papers). Overall fill rate: 63.35%.

Field Category	Hit %	Filled / Total
Synthesis method	85.0%	170 / 200
Crystallinity	83.0%	166 / 200
Building blocks	80.5%	161 / 200
Linkage	69.0%	138 / 200
Solvent stability	69.0%	138 / 200
FT-IR	67.5%	135 / 200
Pore size	63.0%	126 / 200
PXRD pattern	60.5%	121 / 200
SBET	48.5%	97 / 200
Pore volume	7.5%	15 / 200
Overall Fill Rate	1267 / 2000 = 63.35%	

Table 15: ChatExtract domain-specific coverage summary ($N = 200$ papers). Domain-specific coverage: 71.58%.

Field Category	Hit %	Filled / Total
Synthesis method	85.0%	170 / 200
Crystallinity	83.0%	166 / 200
Building blocks	80.5%	161 / 200
Linkage	69.0%	138 / 200
Solvent stability	69.0%	138 / 200
FT-IR	67.5%	135 / 200
Pore size	63.0%	126 / 200
PXRD pattern	60.5%	121 / 200
SBET	48.5%	97 / 200
Pore volume	7.5%	15 / 200
Domain-Specific	859 / 1200 = 71.58%	

C Ablation

C.1 Corpus Growth

Table 16: Phase I ablation (seed refresh): corpus growth. We report the number of *new unique papers* introduced in Round 2 and Round 3, and the cumulative number of unique papers after Round 3. Seed refresh increases late-round novelty and yields a larger expanded corpus.

Setting	Round 2 new	Round 3 new	Total unique (after R3)
No seed refresh	44	29	233
Seed refresh (Set_01)	52	48	260
Seed refresh (Set_02)	45	40	245
Seed refresh (Set_03)	46	47	253